

Assignment 1: Project Understanding and Planning

TASK 1: STUDY THE COMPANY PROBLEM STATEMENT

Problem Statement

The rapid increase in fake news, misinformation, manipulated media, and deepfakes has made it difficult for people to identify trustworthy information. Existing solutions often focus on only one aspect of verification. There is a need for an AI-powered platform that can verify claims, evaluate source credibility, detect deepfakes, and promote digital literacy through explainable AI.

TASK 2: IDENTIFY BUSINESS PROBLEM

Business Problem

The major business problem is the lack of a unified platform for detecting fake news and verifying information. Users often rely on unreliable sources, leading to misinformation, poor decision-making, and reduced public trust. Organizations require an efficient system that provides accurate claim verification and source credibility analysis.

TASK 3: DEFINE PROJECT OBJECTIVES

Objectives

- Detect fake news using AI

Develop an AI-based system that identifies false or misleading news by analyzing the content and comparing it with verified information.

- Verify claims with trusted sources

Check the authenticity of claims using reliable sources such as government organizations, scientific institutions, and fact-checking websites.

- Evaluate source credibility

Analyze whether the source of information is trustworthy based on its reputation and reliability.

- Detect deepfake content

Identify manipulated images, videos, or audio created using AI to prevent the spread of fake media.

- Improve digital literacy and critical thinking

Help users understand how to identify misinformation and encourage responsible use of online information.

TASK 4: SDG MAPPING

SDG 4: Quality Education

Kalki-Kit supports SDG 4: Quality Education by promoting digital literacy, encouraging critical thinking, and providing users with access to accurate and trustworthy information.

TASK 5: IDENTIFY AI OPPORTUNITIES

AI Opportunities

- AI-based Claim Verification

Uses AI algorithms to compare user claims with verified facts and determine their authenticity.

- Deepfake Detection

AI analyzes images, videos, and audio to identify manipulated or AI-generated content.

- Source Credibility Scoring

Assigns credibility scores to information sources based on their reliability and authenticity.

- Explainable AI

Provides clear explanations of why a claim is verified or false, making AI decisions transparent.

- Fake News Prediction and Analysis

Uses AI and data analysis techniques to identify misinformation patterns and predict fake news.

TASK 6: PREPARE FEATURE LIST

Feature List

- AI Claim Verification

Allows users to verify whether a news claim is true or false.

- Source Credibility Evaluation

Checks how trustworthy the source of information is.

- Cross-Reference Engine

Compares claims with multiple trusted databases and sources for accurate verification.

- Fake News Detection Dashboard

Displays verification results, confidence scores, and detailed analysis.

- Deepfake Detection

Detects AI-generated or manipulated images, videos, and audio.

- Social Media Monitoring

Tracks misinformation trends across various social media platforms.

- Evidence Repository

Stores verified facts and supporting evidence for future reference.

- Educational Learning Portal

Provides educational resources to improve awareness about fake news.

- Analytics Dashboard

Shows statistics and graphical insights related to misinformation detection.

- Reports and Notifications

Generates verification reports and sends alerts about misinformation.

TASK 7: DEFINE PROJECT SCOPE

Scope of the Project

The project focuses on developing an AI-powered platform that verifies news, evaluates source credibility, detects deepfakes, and provides educational resources. It is mainly designed for students, educators, researchers, journalists, fact-checkers, government organizations, and NGOs to promote reliable information and reduce misinformation.

TASK 8: PREPARE PROJECT CHARTER

Project Title

Kalki-Kit

Domain

Artificial Intelligence, Fake News Detection, Cybersecurity, Education

Stakeholders

Students: Verify information for assignments and research.

Teachers: Promote digital literacy and fact-based learning.

Journalists: Verify news before publishing.

Researchers: Study misinformation patterns and trends.

Fact Checkers: Improve verification accuracy and efficiency.

Government Organizations: Reduce misinformation and increase public awareness.

NGOs: Support awareness campaigns against fake news.

Overview

Kalki-Kit is a smart AI-powered platform that helps users verify news, detect misinformation and deepfakes, evaluate source credibility, and improve digital literacy. It combines AI, fact-checking, and educational tools to provide accurate and trustworthy information while supporting SDG 4: Quality Education through informed decision-making and responsible digital citizenship.